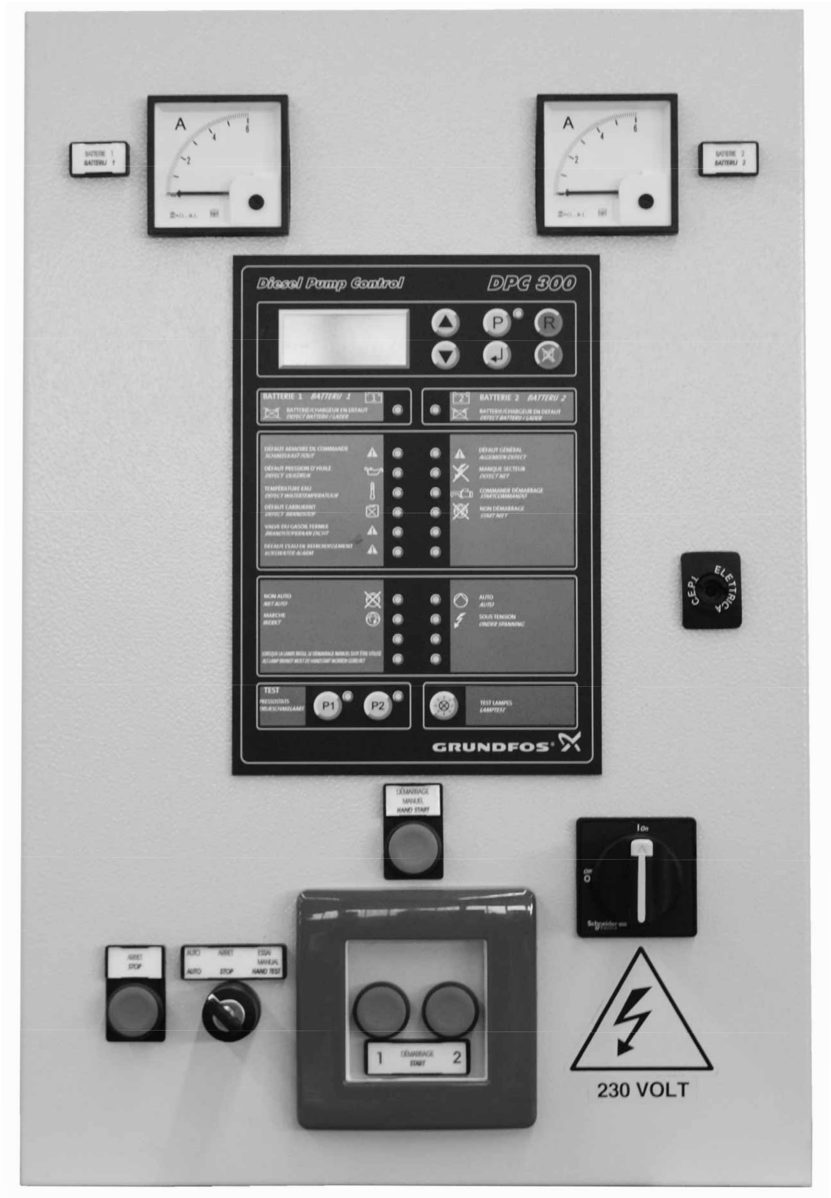
Fire Control EN high diesel 12 & 24 V

Installation and operating instructions



Original installation and operating instructions.

CONTENTS

Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document

Warning

If these safety instructions are not observed, it may result in personal injury.

Warning

If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

Note

Notes or instructions that make the job easier and ensure safe operation.

2. General information

These installation and operating instructions apply to EN 12845 high spec Non-listed and ANPI-approved controllers for fire pump sets driven by a diesel engine with an operating voltage of 12 or 24 V.

Therefore, EN 12845 or ANPI regulations regarding installation, operation and maintenance must be observed when the controllers are integrated into pump sets.

These installation and operating instructions are to be used in conjunction with the following documentation:

- wiring diagram for the controller
- installation and operating instructions for the pump
- installation and operating instructions for the diesel engine
- installation and operating instructions for the pressure switch
- service instructions for the controller.

3. Applications

The controller is designed for automatic operation of pump sets driven by a diesel engine. It is used for automatic start of the pump and monitoring of the diesel engine as well as for manual start of the pump at startup and during maintenance work.

The controller must only be used for control of fire pump sets supplied by Grundfos.

Warning

The controller must only be used for the applications mentioned in these installation and operating instructions. Other applications are considered non-approved. Grundfos cannot be held responsible for possible consequential damage. The risk is carried solely by the operator.

The control cabinet must not be used to supply voltage to other pump systems, not even other fire pump sets.

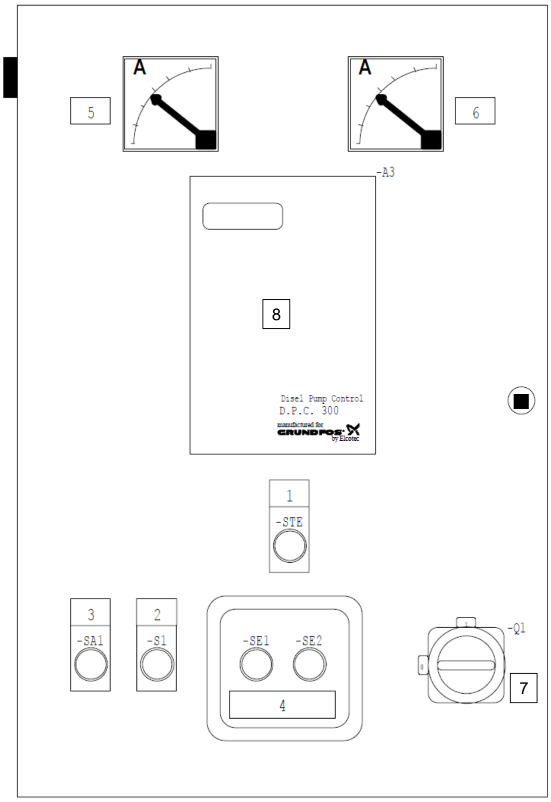
4. Product description

The controller is placed in a control cabinet with fixing holes at the back and in the bottom for mounting. The cabinet can either be mounted on a pump set or on a wall.

The control cabinet, which is powered by the mains, supplies the engine with the power required for starting and operating the engine. Furthermore, the power supply is used to charge two battery sets. Each battery set is charged via its own charger.
If the power supply fails, the two battery sets take over the electrical supply of the control cabinet.

4.1 Controls and indicating instruments

All controls and indicating instruments are placed in the cabinet door. See fig. 1.



TM05 4957 3012

Fig. 1 Cabinet door

Pos.	Design.	Description
1	-	Control panel
2	STE	Manual start switch
3	SA1	Stop switch
4	S1	Lockable selector switch (AUTO - STOP - MANUAL TEST)
5	SE1	Button for emergency start 1
6	SE2	Button for emergency start 2
7	Q1	Main switch
8	-	Ammeter for battery set 1
9	-	Ammeter for battery set 2
10	-	Buzzer

The control elements and indicator lights have the following functions:

Control panel (DPC 300) (pos. 1)

The control panel contains the buttons required to set the parameters of the controller, a text display to display status and fault messages as well as various indicator lights. It is placed in the cabinet door. The control panel is described in section *8. Control panel*.

Manual start switch (pos. 2)

The button is used to start the engine manually if it does not start after six automatic start attempts. It is also used in connection with test running and startup. You can only activate the button when the selector switch is set to "MANUAL TEST". See section *12.3 Emergency operation*. The two battery sets will be used alternately.

Stop switch (pos. 3)

Whether started manually or automatically, the diesel engine can be stopped by means of [STOP]. See sections *12.1 Automatic operation* and *12.2 Manual operation*. You can only activate the button when the selector switch is set to "MANUAL TEST".

Selector switch (pos. 4)

The selector switch is used to select the operating mode (manual or automatic). Remove the key from the selector switch to prevent unauthorised activation.

Buttons for emergency start (pos. 5 and 6)

If the controller fails, the engine can still be activated via the two buttons behind the glass panel. See section *12.3 Emergency operation*. Each battery set has its own button.

Main switch (pos. 7)

The main switch is used to switch on the power supply to the control cabinet.

Ammeter (pos. 8 and 9)

The two ammeters indicate the charging current of the two battery sets. Each battery set has its own ammeter.

Buzzer (pos. 10)

The buzzer is activated in case of an alarm situation.

4.2 Main components in control cabinet

The main components in the control cabinet are shown in fig. 2.

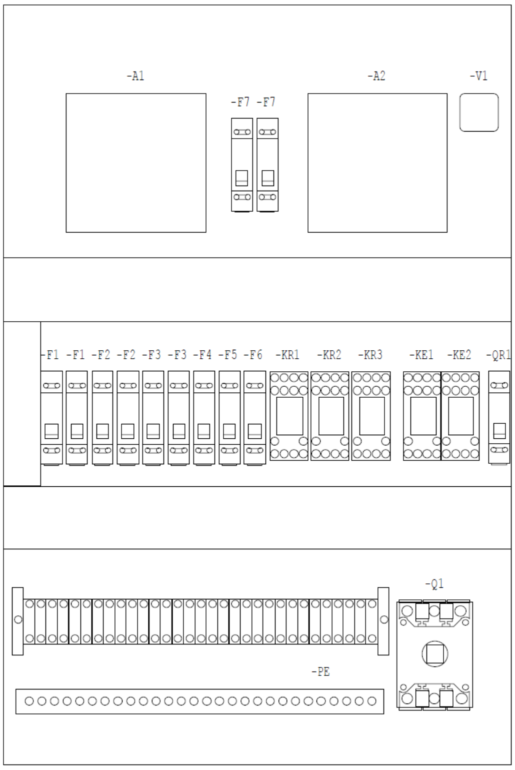


Fig. 2 Main components in control cabinet

Pos.	Description
A1	Charger 1
A2	Charger 2
F1	Main fuse
F2	Fuse, charger 1
F3	Fuse, charger 2
F4	Fuse, battery set 1
F5	Fuse, battery set 2
F6	Fuse, controller
F7	Fuse, emergency activation via battery set
KE1	Changeover relay, battery set 1
KE2	Changeover relay, battery set 2
KR1	Control relay, mains voltage available
KR2	Relay, activation of engine block heater in automatic mode
KR3	Relay, activation of engine block heater in manual mode
PE	Protective conductor bar
Q1	Main switch
QR1	Automatic circuit breaker, preheating of cooling water
V1	Diode bridge

Inputs and outputs

The inputs and outputs are listed in section *7. Technical data*.

5. Functions

5.1 Operating functions

Automatic operation

Note | ***Automatic operation is the normal operating mode. For this mode, the selector switch 1 must be set to "AUTO".***

When the pump set is in automatic mode, only the green "SUPPLY ON" indicator light is on. All other alarm and status indicator lights should be off.

As soon as the sprinklers are activated and water is consumed, the pressure in the discharge pipe is reduced. If the pressure falls below the cut-in pressure set on the pressure switch, the pump will start automatically. The red "PUMP RUNNING" indicator light will light up, "AUT." will be displayed as the reason for the startup in accordance with the position of the selector switch, and the AR1 digital output will be activated. During startup, the number of startup attempts and the currently used starting unit (battery set 1 or 2) are also shown in the display. The green "AUTO" indicator light is on.

A start attempt lasts five to ten seconds. If the engine does not start, another start attempt will be made after five to ten seconds. The duration of start attempts and the time interval between two start attempts are adjustable. See section *8.2 Settings*.
A maximum of six start attempts will be made. After every failed start attempt, the controller changes over to the other battery set. After six failed start attempts, the controller blocks the automatic operation, and the yellow "START FAILURE" indicator light in the control panel lights up. The message "Start failure" appears in the display, and the AR4 digital output is activated. Press [R] to reestablish automatic operation. See section *12. Operation*.

It is only possible to stop the pump manually by setting the selector switch to "MANUAL TEST" and pressing [STOP]1.

For safety reasons, a redundant pressure switch should always be connected i addition to the primary pressure switch.

See section *12.1 Automatic operation*.

Manual operation

During startup and for test purposes, the pump can be started manually by setting the selector switch to "MANUAL TEST", thus bypassing the pressure switch in the discharge line.
This launches the automatic start process as described under "Automatic operation". The yellow "NON AUTO" indicator light and the red "PUMP RUNNING" indicator light will light up, "TEST" will be displayed as the reason for the startup in accordance with the position of the selector switch, and digital outputs AR1 and AR3 will be activated.

See section *12.2 Manual operation*.

Emergency start function

The controller is equipped with an emergency start device in case the pump does not start automatically. The yellow "OPERATE MANUAL START TEST BUTTON IF LAMP IS LIT" indicator light indicates whether the pump can be started by means of [MANUAL START]. The indicator light is on in these cases:

- when the controller has been switched on for the first time
- when the engine has been started automatically and subsequently stopped by means of [STOP]
- after six failed start attempts.

The indicator light goes off when the engine is switched off.

How to test the emergency startup device is described in section *12.4 Test run*.

In case of controller fault (the "DIESEL CONTROLLER FAILURE" indicator light is on, or the display and the indicator lights are off), break the glass and start the pump with [START 1] or [START 2].

Both emergency start functions are described in section *12.3 Emergency operation*.

Pressure switch test

The function of the pressure switches can be tested by means of [P1] and [P2]. To do this, set the selector switch to "MANUAL TEST", and press [P1] for pressure switch 1 or [P2] for pressure switch 2. During the testing procedure, the indicator light next to the button that has been pressed will light up.
See section *12.4 Test run*.

Stop function

When the fire pump set is switched off by means of [STOP], the controller interrupts the fuel supply. Depending on the design, it is interrupted by switching off the power supply to the diesel injection pump or via a valve (energized to stop).

Calibration function

During startup, the engine speed must be calibrated. This will ensure that the correct speed is always shown in the display.
See section *8.2.3 Engine speed*.

Louvre control

To ensure a sufficient supply of fresh air to the engine and targeted heat removal, it may be necessary to install louvres. The louvres can be controlled via the control cabinet.
The controller or, in case of power failure, the battery sets power the motors of the louvres. The louvres are opened automatically with the "PUMP RUNNING" indication and closed when the engine is stopped. It is only possible to connect louvres to 230 V.

Indicator light test

It is possible to carry out an indicator light test. To test the function of the indicator lights of the control panel, press [⊗].

Engine block heater

Some pump sets are equipped with a engine block heater for the engine. The control cabinet contains a 10 A automatic circuit breaker and the required terminals for power supply to the engine block heater. The heater adjusts itself via an external cooling water thermostat.

5.2 Monitoring functions

The most important fault indications can, via a digital output, be forwarded to a building management system as single indications or a common alarm. The following faults are combined to one common alarm:

- no mains voltage
- defective battery/charger
- oil pressure too low
- engine temperature too high
- fuel tank filling level too low
- engine block heater fuse has tripped.

The engine is not switched off in case of faults.

See section *8.3 Status and fault indications*.

Phase failure

The power supply to the control cabinet is monitored by a relay. As long as the power supply is within the permitted range, the "SUPPLY ON" indicator light is on. If the mains supply fails for more than 180 seconds, the "SUPPLY ALARM" indicator light lights up and the message "Power supply" appears in the bottom line of the display. The AR2 digital output is activated. When the fault no longer exists, the fault indication will be reset automatically.

Battery voltage

Both battery sets are monitored. If the battery voltage is less than 21 V or more than 29 V for more than 30 seconds, or no battery is connected, the warning message "Fault, battery 1" or "Fault, battery 2" will appear in the display for the battery concerned, and the yellow "BATTERY 1 -> BATTERY/BATTERY CHARGER FAILURE" or "BATTERY 2 -> BATTERY/BATTERY CHARGER FAILURE" indicator light will light up. The AR2 digital output will also be activated. The fault indication must be reset manually by pressing [R].

Charging voltage

Both chargers are monitored. If a charger does not work properly for more than 10 seconds, the warning "Fault, charger 1" or "Fault, charger 2" will appear in the display for the charger concerned and the yellow "BATTERY 1-> BATTERY/BATTERY CHARGER FAILURE" or "BATTERY 2 -> BATTERY/BATTERY CHARGER FAILURE" indicator light will flash. The AR2 digital output will also be activated. When the fault no longer exists, the fault indication will be reset automatically.

Motherboard

The motherboard of the controller is monitored. In case of failure, the yellow "DIESEL CONTROLLER FAILURE" indicator light will light up in the control panel. Furthermore, the AR5 digital output will be activated, which means it is still possible to start the fire pump set by bypassing the controller via [START 1] and [START 2] behind the glass panel.

Selector switch position

The fire pump set is only ready for fire fighting when the selector switch is set to "AUTO". This is why the position of the selector switch is monitored. If the selector switch is not set to "AUTO", but to "MANUAL TEST" or "STOP", the yellow "NON AUTO" indicator light will be on and the AR3 digital output activated.

Pump start/speed sensor

Whether the pump is running or not is monitored via a signal from the speed sensor and an optional pressure switch (installed in the discharge pipe). If there is a signal from the speed sensor and the pressure switch detects pressure, the "PUMP RUNNING" indicator light will light up, and the AR1 digital output will be activated. Furthermore, the engine speed will appear in the display.

If there is no signal from the speed sensor, "Stop" will be displayed instead of the speed. If the "Stop" message does not disappear after pressing [R], the speed sensor is defective. If the contact of the pressure switch is open (no pressure), the "PUMP RUNNING" indicator light will be off, and the AR1 digital output will not be activated.

The sensor connection of the additional pressure switch can also be monitored for wire breakage and short-circuit. In case of sensor cable fault, the alarm message "Alarm, pump on" will be shown in the display, and the AR2 digital output will be activated. When the fault no longer exists, the fault indication will be reset automatically.

If no pressure switch is connected, terminals 93 and 94 must be short-circuited.

Oil pressure

The oil pressure is monitored via a pressure switch on the diesel engine. The monitoring function is activated 60 seconds after the engine has been started. If the oil pressure becomes too low, the "LOW OIL PRESSURE" indicator light will light up after a short delay. The display shows the message "Low oil press.", and the AR2 digital output is activated. The engine will not be stopped in case of low oil pressure. The fault indication must be reset manually by pressing [R].

The sensor connection of the pressure switch can also be monitored for wire breakage and short-circuit. In case of sensor cable fault, the alarm message "Alarm, oil press." will be shown in the display, and the AR2 digital output will be activated. When the fault no longer exists, the fault indication will be reset automatically.

Engine temperature

The temperature is monitored via a thermal switch on the engine. The monitoring function is activated 10 seconds after the engine has been started. If the temperature of the engine cooling liquid becomes too high, the "ENGINE OVERHEATED" indicator light will light up after a short delay. The display shows the message "Engine temperat.", and the AR2 digital output is activated. The engine will not be stopped in case the temperature becomes too high. The fault indication must be reset manually by pressing [R].

The sensor connection of the thermal switch can also be monitored for wire breakage and short-circuit. In case of sensor cable fault, the alarm message "Engine temperat." will be shown in the display, and the AR2 digital output will be activated. When the fault no longer exists, the fault indication will be reset automatically.

Cooling water flow

The flow in the cooling water circuit is monitored by a flow switch. If there is no flow, the yellow "COOLING WATER ALARM" indicator light will light up, and the AR2 digital output will be activated. The engine will not be stopped in case there is no flow in the cooling water circuit. The fault must be reset manually by pressing [R]

Fuel tank level

The fuel level in the diesel fuel tank is monitored.

If the level in the fuel tank falls below 70 %, the "FUEL STOCK" indicator light will light up. The display shows the message "Fuel reserve", and the AR2 digital output is activated. When the level increases to above 75 %, the fault indication will be reset automatically.

Fuel valve position

The position of the fuel valve is monitored. When the fuel valve is closed, the contact at the inlet is open. The display shows the alarm message "Valve closed", and the yellow "VALVE CLOSED" indicator light is on. The AR2 digital output will also have been activated. When the fault no longer exists, the fault indication will be reset automatically.

The sensor cable for monitoring the fuel valve position can also be monitored for wire breakage and short-circuit. In case of sensor cable fault, the alarm message "Alarm, valve closed" will be shown in the display, and the AR2 digital output will be activated. When the fault no longer exists, the fault indication will be reset automatically.

Liquid level in priming tank

The level in the priming tank can be monitored by installing a level switch. If the level falls under a certain value, for instance because of a leakage in the system, the engine will be started. Digital outputs AR1 and AR2 will be activated. When the fault no longer exists, the fault indication will be reset automatically.
A priming tank must be provided if there is a negative inlet pressure or in other cases defined in the EN 12845 or ANPI regulations.

The sensor cable for starting the pump set by means of the level switch in the priming tank can also be monitored for wire breakage and short-circuit. In case of sensor cable fault, the alarm message "Alarm, tank level" will be shown in the display, and the AR2 digital output will be activated. When the fault no longer exists, the fault indication will be reset automatically.

Optional engine block heater

The fuse of the engine block heater is monitored. If the fuse trips, the "COOLING WATER ALARM" indicator light will light up.
The display shows the message "Short-c. heater", and the AR2 digital output is activated. When the fault no longer exists, the fault indication will be reset automatically.

External devices

It is possible to monitor an external device, such as a shut-off valve, via the controller. Use the DI11 digital input for this. If the contact is closed, the alarm message "External fault" will appear in the display. The fault indication must be reset manually by pressing [R].

6. Identification

The controller can be identified by means of the nameplate on the right side of the device and the inside of the cabinet door.

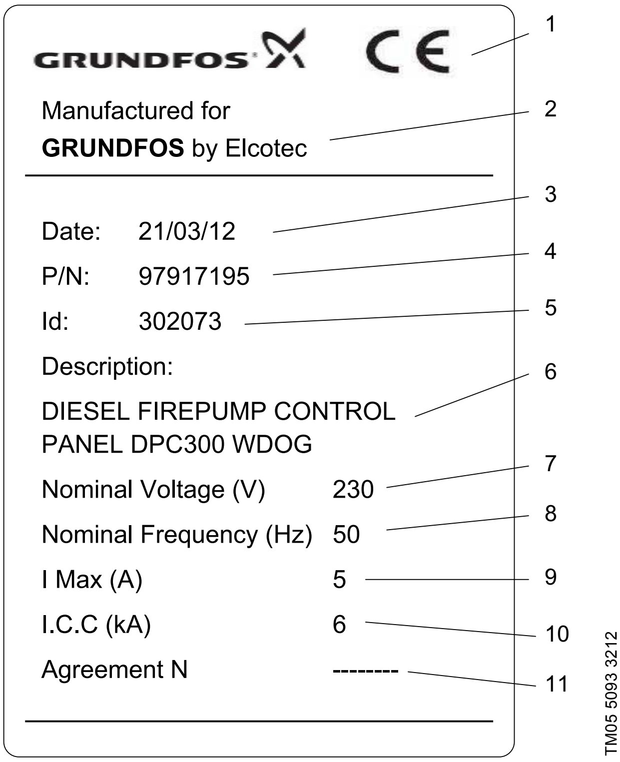


Fig. 3 Nameplate

Pos.	Description
1	CE mark
2	Manufacturer
3	Production date
4	Product number
5	Serial number
6	Type designation
7	Rated voltage
8	Rated frequency
9	Max. current
10	Rated short-circuit current
11	Options

7. Technical data

Control cabinet:	Sheet steel, grey.
Enclosure class:	IP44.
Dimensions w x h x d:	400 x 600 x 250 mm.
Weight:	Approx. 50 kg.
Supply voltage:	1 x 230 V, 50 Hz, N + PE.
Max. power consumption:	100 W without engine block heater. 1100 W with engine block heater.
Max. current consumption:	10 A.
Connecting terminals:	L1, N = 2.5 - 6 mm^2.
EMC noise emission:	According to EN 61000-6-3.
EMC immunity:	According to EN 61000-6-2.
Overvoltage category:	III.
Rated surge voltage:	6 kV.
Rated short-circuit current:	6 kA.
Degree of contamination:	2.
Ambient temperature:	0-50 °C.
Country of origin:	Italy.

Inputs

The controller has four analog inputs and seven digital inputs. See tables below. The tables also state terminal designations and, for digital inputs, the status when activated.

Analog inputs

Desig nation	Analog input	Terminal	Range of values	Corresponding resistance [Ω]
AI1	Fuel level	63	0-100 %	350-0
AI2	Engine temperature	65	40-150 °C	1064-34
AI3	Oil pressure	64	0-80 bar	270-0
AI4	Speed sensor	61 + 62	-	-

Electrical data of analog inputs

Voltage class:	None.
Insulation voltage:	50 V (in relation to earth).
Insulation test voltage:	None.
Input resistance:	1.5 kOhm.
Input resolution:	19.5 mV.
Accuracy at 25 °C:	± 0.4 %.
Resolution:	8 bits.
Integral time:	150 ms.
Reading interval:	40 ms.

The analog inputs are not short-circuit-proof.

Data of the pulse input to which the speed sensor can be connected for monitoring of the generator:

Max. frequency:	10 kHz.
Input resistance:	4.85 kOhm.
Input voltage:	± 30 V.
Input current:	7 mA.

Digital inputs

Designation	Digital input	Terminal	Position when activated
DI1*	Pressure switch 1	81 + 82	Open
DI2*	Pressure switch 2	83 + 84	Open
DI3*	Level switch, priming tank (to start the engine)	85 + 86	Closed
DI4*	Oil pressure switch (low oil pressure)	89 + 90	Closed
DI5*	Thermostat (high engine temperature)	91 + 92	Closed
DI6*	Pressure switch (pump running)	93 + 94	Closed
DI7*	Fuel valve position	87 + 88	Closed
DI8	Flow switch (no flow in cooling circuit)	70 + 95	Closed
DI9	Remote alarm heater	96 + 97	Open
DI10	Control of louvres	98 + 99	
DI11	Auxiliary input 1	70 + 100	Closed

* Can be monitored.

The digital inputs are not short-circuit-proof.

Outputs

The controller has six digital outputs which are potential-free changeover contacts. The signals from the digital outputs can be passed on to a building management system.

Digital outputs

Designation	Digital outputs	Terminal
AR1	Pump running	30-33
AR2	Common alarm	34-37
AR3	Selector switch	38-40
AR4	Start failure	41-43
AR5	Controller fault	44-46
AR6	Auxiliary output 1	47-49

Note ***Auxiliary output 1 is not wired internally and consequently not used.***

Electrical data of signal relay outputs

Voltage class:	Category 1.
Insulation voltage:	115 V (in relation to earth).
Insulation test voltage:	1.5 kVAC.
Max. supply voltage:	115 VAC.
Max. load:	2 A, 115 V.
Min. load:	100 mA, 12 VDC.
Max. load power:	230 VA / 24 W.

7.1 Factory settings

See section *8.2 Settings* for information on setting the controller.

8. Control panel

The control panel is used to set parameters, to display status and alarm indications via the display and the indicator lights as well as to reset alarms.

The control panel is divided into two parts (see fig. 4):

- the display with operating buttons
- the status and fault indicator lights.

The indicator light part is subdivided into these parts:

- fault indications, battery set 1 and 2
- alarm indications
- status indications
- pressure switch and lamp testing.

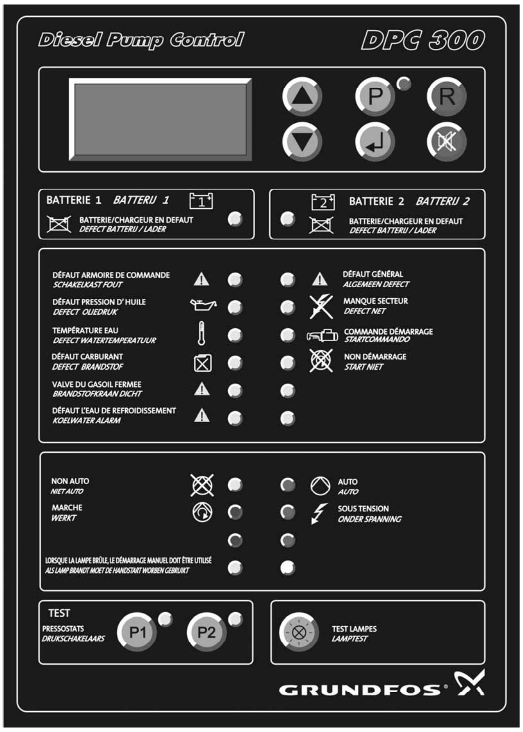


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Fig. 4 Control panel

Display

The display shows the status via operating data and warning messages as well as the individual setting displays.

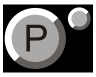
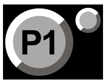
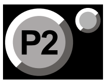
The setting displays are described in section *8.2 Settings*, and the warning messages shown in the bottom line are described in section *8.3.2 Fault indications*. Figure 5 shows the design of the status display.

1700 rpm	AUT	>	Speed or calibration prompt or battery set and number of start attempts (to the left) and the cause for starting (to the right).
T = 40 °C	Oil = 10b	>	Engine temperature (°C) and oil pressure (bar)
00000H	Fuel = 100 %	>	Operating hours (H) and fuel level (%)
B1 = 24 V	B2 = 24 V	>	Voltage of battery set 1 and 2 or warning message in case of faults

Fig. 5 Example of status display

Button functions

The following table shows the function of the individual operating buttons.

Button	Function	Symbol in text
 	Setting of contrast Navigation between displays Changing of parameter values	[▲] [▼]
	Entering selection mode Exiting selection mode	[P]
	Activation of entry mode Storing of parameter values	[↵]
	Resetting of alarms	[R]
	Resetting of buzzer	-
	Indicator light test	[⊗]
	Test of pressure switch 1	[P1]
	Test of pressure switch 2	[P2]

Indicator lights

The indicator lights are explained in section *8.3 Status and fault indications*.

8.1 Menu structure

The menu structure including the sequence of setting displays is illustrated below.

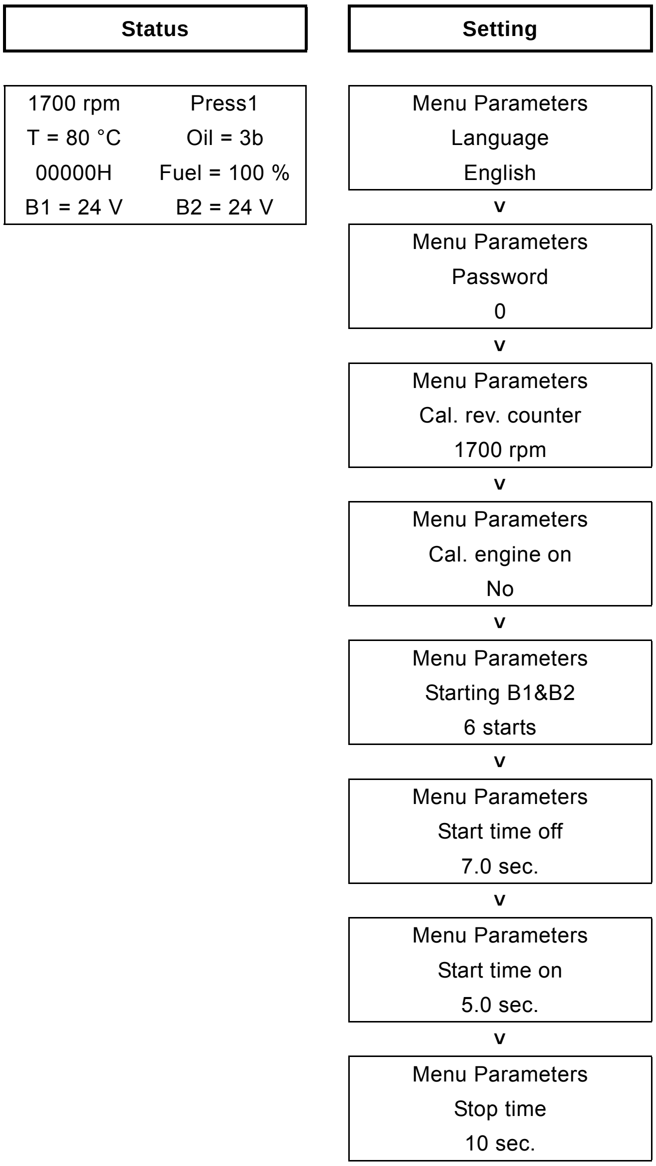


Fig. 6 Menu structure

8.2 Settings

To change parameters, proceed as follows:

1. Press [P] to enter the selection mode. The green indicator light next to [P] lights up, and instead of the status display the first setting display will be shown.
2. Select the setting display of the parameter you wish to change with [▲] or [▼].
3. Press [↵] to change the parameter. The green indicator light next to [P] starts flashing.
4. Change the parameter value by pressing [▲] or [▼].
5. Press [↵] to save the parameter. The green indicator light next to [P] is now on.
6. To change other parameters, repeat steps 2 to 5.
7. Press [P] to activate the changes and return to the status display.

8.2.1 Language

Menu Parameters
Language
German

In the "Language" display, you can select one of the following languages:

- English
- French
- Dutch.

Factory setting: Local language, if available, or English.

8.2.2 Password

Menu Parameters
Password
0

In this display, a password can be entered in order to change further parameters. These parameters may only be changed by Grundfos employees.

8.2.3 Engine speed

If the controller is part of a complete fire pump set, the speed display will have been calibrated at the factory. If the controller is delivered separately, however, a request for speed calibration will appear in the first line of the status display during the startup:

?CAL?	
T = 40 °C	Oil = 10b
00000H	Fuel = 100 %
B1 = 24 V	B2 = 24 V

Calibration procedure:

1. Measure the operating speed directly at the diesel engine with a suitable measuring device. Set the operating speed of the engine to the next full hundred step (e.g. 1700 rpm). Setting of operating speed is described in the installation and operating instructions for the diesel engine.
2. Press [P] to enter the selection mode. The green indicator light next to [P] lights up.
3. Select the following setting display using [▲] or [▼]:

Menu Parameters
Cal. rev. counter
1700 rpm

4. Press [↵] to change the speed. The green indicator light next to [P] starts flashing.
5. Adjust the speed by pressing [▲] or [▼] (setting range 1000 to 4000 rpm, adjustable in steps of 100).
6. Save the parameter value by pressing [↵]. The green indicator light next to [P] is now on.
7. Press [P] to activate the changes and return to the status display.

After the automatic calibration, the parameter value in the "Cal. engine on" display is automatically reset to "No":

Menu Parameters
Cal. engine on
No

The "?CAL?" symbol flashes in the display during the automatic calibration process. The calibration process can take up to 30 seconds. When the calibration process is completed, the current speed appears instead of the flashing symbol.

8. After calibration, the operating speed of the diesel engine must be reset to the original nominal speed.

8.2.4 Number of start attempts

Menu Parameters
Starting B1&B2
6 starts

This display shows the number of start attempts. It is not possible to change this value.

Factory setting: 6.

8.2.5 Start attempt interval

Menu Parameters
Start time off
7.0 sec.

In this display, the interval between two start attempts can be set for automatic operation.

Setting range: 5.0 to 10.0 seconds in steps of 0.1 second.

Factory setting: 7.0 seconds.

Note ***In order to speed up the setting procedure, the parameter value can also be changed in steps of 1.0 seconds by keeping [▲] or [▼] pressed.***

8.2.6 Duration of start attempts

Menu Parameters
Start time on
5.0 sec.

In this display, the number of seconds for which the starting relay is activated in order to start the diesel engine can be set for automatic operation. The duration of the start attempt is thus specified.

Setting range: 5.0 to 10.0 seconds in steps of 0.1 second.

Factory setting: 5.0 seconds.

Note ***In order to speed up the setting procedure, the parameter value can also be changed in steps of 1.0 seconds by keeping [▲] or [▼] pressed.***

8.2.7 Duration of stop attempts

Menu Parameters
Stop time
10 sec.

In this display, you can set the number of seconds for which the stop valve is activated in order to stop the engine when [STOP] is pressed.

Setting range: 5.0 to 25.0 seconds in steps of 0.1 second.

Factory setting: 10 seconds.

Note ***In order to speed up the setting procedure, the parameter value can also be changed in steps of 1.0 seconds by keeping [▲] or [▼] pressed.***

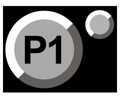
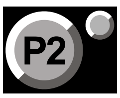
8.2.8 Contrast

The display contrast can be adjusted by pressing [▼] (darker) or [▲] (brighter) in the status display.

8.3 Status and fault indications

8.3.1 Status indications

The status is indicated as text in the display (see section *8. Control panel*) and by means of indicator lights in the control panel. The following table provides an overview of status indications.

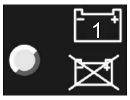
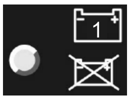
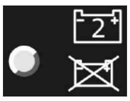
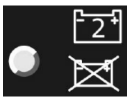
Status	Indicator light			Display text (status display)	Output activated
		Status	Text		
The pump set is in automatic mode and can start as required via the pressure switch or the level switch in the priming tank. The selector switch is set to "AUTO".		On	AUTO	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % B1 = 24 V B2 = 24 V	
The pump is running (automatic start via pressure switch or manual start). Whether the pump was been started manually or automatically is shown in the top line of the display where the position of the selector switch is indicated to the right: AUT. = automatic operation TEST = manual operation STOP = switched off.		On	PUMP RUNNING	1700 rpm AUT T = 80 °C Oil = 3b 00000H Fuel = 100 % B1 = 24 V B2 = 24 V	AR1
Mains voltage correct.		On	POWER SUPPLY	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % B1 = 24 V B2 = 24 V	
Calibration prompt (at startup).	-	-	-	?CAL? T = 20 °C Oil = 0b 00000H Fuel = 100 % B1 = 24 V B2 = 24 V	
Engine start during automatic operation. Indication of battery set (B1), number of start attempts (3) and start trigger (AUT). (Between starts, the message "Engine off" is shown in the top line of the display.)		On	ON DEMAND	Star. B1 = 3 AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % B1 = 24 V B2 = 24 V	
Engine start during manual operation. (When the engine has started, the speed is indicated in the upper left corner of the display.)	-	-	-	Engine off TEST T = 20 °C Oil = 0b 00000H Fuel = 100 % B1 = 24 V B2 = 24 V	
This yellow status indicator light lights up • when the controller has been switched on for the first time • when the engine has been started automatically and subsequently switched off by means of [STOP] • after six failed start attempts. In these situations, it is possible to start the pump set by pressing the [MANUAL START].		On	OPERATE MANUAL START TEST BUTTON IF LAMP IS LIT	Engine off TEST T = 20 °C Oil = 0b 00000H Fuel = 100 % B1 = 24 V B2 = 24 V	
Test of pressure switch 1 is running.		On	PRESSURE SWITCHES	Engine off TEST T = 40 °C Oil = 3b 00000H Fuel = 100 % B1 = 24 V B2 = 24 V	
Test of pressure switch 2 is running.		On	PRESSURE SWITCHES	Engine off TEST T = 40 °C Oil = 3b 00000H Fuel = 100 % B1 = 24 V B2 = 24 V	

8.3.2 Fault indications

Faults are indicated as warning messages in the bottom line of the display and by means of indicator lights in the control panel or indicator lights in the cabinet door. Some warnings are stored and must be reset manually by pressing [R], even if the fault is no longer present. Other warnings disappear automatically when the fault is no longer present and require no resetting.

At first, the message "*** Alarm ***" is shown in the bottom line of the display. Then the display with the warning message is shown. After a certain interval (a few seconds), the message "*** Alarm ***" is shown again, followed by the next warning message.

In case of warning messages that are both shown for pumps ready for operation and pumps in operation, the table shows the display of a pump ready for operation.

Fault	Indicator light			Display text (status display)	Output activated	Resetting
		Status	Text			
This indicator light lights up when one or more of the indicator lights described below is/are activated.		On	GENERAL ALARM			
Phase failure, no power supply.		On	SUPPLY ALARM	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % Power supply		Automatic
The fire pump set is not in automatic mode. The selector switch is set to "STOP" or "MANUAL TEST".		On	NON AUTO	Engine off STOP T = 20 °C Oil = 0b 00000H Fuel = 100 % Selector switch: STOP	AR3	
Battery set 1: Battery voltage < 21.0 V Battery voltage > 29.0 V No battery connected.		On	BATTERY 1 BATTERY/BATTERY CHARGER FAILURE	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % Fault, battery 1	AR2	Manual
Charger 1: Low or no charging voltage.		Flashing	BATTERY 1 BATTERY/BATTERY CHARGER FAILURE	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % Fault, charger 1	AR2	Automatic
Battery set 2: Battery voltage < 21.0 V Battery voltage > 29.0 V No battery connected.		On	BATTERY 2 BATTERY/BATTERY CHARGER FAILURE	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % Fault, battery 2	AR2	Manual
Charger 2: Low or no charging voltage.		Flashing	BATTERY 2 BATTERY/BATTERY CHARGER FAILURE	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % Fault, charger 2	AR2	Automatic
Controller failure.		On	DIESEL CONTROLLER FAILURE	-	AR5	
Oil pressure too low (only when engine is running).		On	LOW OIL PRESSURE	1700 rpm AUT T = 80 °C Oil = 1b 00000H Fuel = 100 % Low oil press.	AR2	Manual

Fault	Indicator light			Display text (status display)	Output activated	Resetting
		Status	Text			
Cooling water temperature too high (only when engine is running).		On	ENGINE OVERHEATED	1700 rpm AUT T = 100 °C Oil = 3b 00000H Fuel = 100 % Engine temperat.	AR2	Manual
Low level in fuel tank (level < 70 %).		On	FUEL STOCK	Engine off AUT T = 20 °C STOP = 0b 00000H Fuel = 20 % Fuel reserve	AR2	Automatic
Fuel valve closed.		On	VALVE CLOSED	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % Valve closed	AR2	Automatic
No flow in the cooling circuit of the engine.		On	COOLING WATER ALARM	1700 rpm AUT T = 100 °C Oil = 3b 00000H Fuel = 100 % Cooling water	AR2	Manual
Six failed start attempts (automatic mode).		On	DIESEL FIRE PUMP FAILURE TO START	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % Start failure	AR4	Manual
Connection to pressure switch 1 defective.	-	-	-	Engine off T = 20 °C Oil = 0b 00000H Fuel = 100 % Alarm, press. 1	AR2	Manual
Connection to pressure switch 2 defective.	-	-	-	Engine off T = 20 °C Oil = 0b 00000H Fuel = 100 % Alarm, press. 2	AR2	Manual
Connection to level switch in priming tank defective.	-	-	-	Engine off T = 20 °C Oil = 0b 00000H Fuel = 100 % Alarm, tank level	AR2	Manual
Connection to low oil level switch defective.	-	-	-	Engine off T = 20 °C Oil = 0b 00000H Fuel = 100 % Alarm, oil press.	AR2	Manual
Connection to thermal switch defective.	-	-	-	Engine off T = 20 °C Oil = 0b 00000H Fuel = 100 % Engine temperat.	AR2	Manual
Connection to pressure switch for "pump running" defective.	-	-	-	Engine off T = 20 °C Oil = 0b 00000H Fuel = 100 % Alarm, pump on	AR2	Manual
Connection of signal cable for "valve closed" defective.	-	-	-	Engine off T = 20 °C Oil = 0b 00000H Fuel = 100 % Valve closed	AR2	Manual

Fault	Indicator light			Display text (status display)	Output activated	Resetting
		Status	Text			
Speed sensor fault (only when engine is running).	-	-	-	STOP AUT T = 80 °C Oil = 3b 00000H Fuel = 100 % Rev. counter sens.		Manual
External device fault.	-	-	-	Engine off AUT T = 20 °C Oil = 0b 00000H Fuel = 100 % External fault		Manual

9. Delivery, transport and storage

Note

Check the controller carefully on delivery, and transport and store it correctly before installation.

If the controller is not already mounted on the pump set, it is delivered from the factory in an open wooden crate on a pallet.

Caution

The controller must only be transported in the packaging. Use only suitable lifting equipment that is in good condition.

In order to prevent damage to the controller, leave the control cabinet in the packaging until installation.

Storage temperature: 0-50 °C.

Relative humidity: 95 %, not condensing.

10. Installation

Warning

The controller must be installed and connected in accordance with local regulations and by qualified personnel.

10.1 Mechanical installation

Warning

Installation and operation in a potentially explosive environment is not permitted.

Warning

The control cabinet must be placed so that water from the pump or pipework cannot pose any danger.

Caution

Use only suitable lifting equipment that is in good condition. Always lift the control cabinet by means of the lifting eyes. Close the cabinet door before transporting the control cabinet.

The controller is only suitable for indoor installation. It must not be exposed to direct sunlight and must be installed in a dry, well-ventilated but frost-free room.

The controller must be placed as close to the pump as possible and within view of the pump set. The maximum cable length is 30 m. Ensure sufficient space for installation and subsequent access to the controller, and make sure that the control cabinet is installed at the correct height. Make sure there is sufficient air circulation for cooling.

There are two mounting options:

1. mounting on the wall
2. mounting on the fuel tank.

Mounting on the wall

Mount the control cabinet on the wall near the pump set by means of screws. Drill holes into the wall according to the drilling template on the back of the control cabinet, and use dowels and screws of adequate size.

Warning

When drilling holes, make sure not to damage any wires or water and gas pipes. Furthermore, ensure safe installation.

Mounting on the fuel tank

Pump sets delivered by Grundfos have a special retainer for the control cabinet on the fuel tank. Mount the control cabinet in the retainer by means of suitable screws and nuts.

10.2 Electrical connection

Warning

During electrical installation, make sure that the power supply cannot be accidentally switched on.

Carry out the electrical connection according to local regulations.

Check that the supply voltage and frequency correspond to the values stated on the nameplate.

Warning

The electrical installation should be carried out by an authorised person in accordance with local regulations and the relevant wiring diagram.

Procedure

- The mains cable must be sized according to the relevant standards, and the installation must be equipped with fuses according to the cable cross-sections. Connect the mains cable to the main switch. PE (protective earth) must be connected using the PE screw.
- Connect pressure switch 1 to terminals 81 and 82 and pressure switch 2 to terminals 83 and 84 in the control cabinet.
- If status and alarm indications are to be passed on to a building management system, use the following digital outputs:

Signal	Terminal
Pump running	30-33
Common alarm	34-37
Selector switch	38-40
Start failure	41-43
Controller fault	44-46

- If there is a level switch for starting the pump when the level in the priming tank falls below a certain level, the level switch must be connected to terminals 85 and 86.
- If louvres are to be controlled and powered by the controller, connect the motor of the louvres to terminals L3 and N. Connect the signal cables to terminals 98 and 99.
- If an engine preheater is required, connect the engine block heater to terminal L2 and N, and the preheater thermostat lead to terminal 96 and 97.

If the controller is delivered separately, e.g. in case of replacement, the following additional connections have to be made as well:

- Connect the cables of the two battery sets. Connect L+ of battery set 1 to terminal 1 and L+ of battery set 2 to terminal 2. Connect the negative poles to the two 0-terminals.
- Connect the stop valve for interruption of the fuel supply to terminal 28 (digital output).
- Connect the cooling solenoid valve to terminal 29.
- Connect the signal cables of the fuel valve to terminals 87 and 88.
- Connect the following sensor cables from the fire pump set to the analog inputs below.

Sensor cable	Terminal
Temperature sensor	65
Oil pressure sensor	64
Level sensor, fuel tank	63
Speed sensor	61 + 62

- Connect the following sensor cables from the fire pump set to the digital inputs below.

Sensor cable	Terminal
Oil pressure switch	89 + 90
Temperature switch, engine	91 + 92
Pressure switch for pump running	93 + 94
Flow switch	70 + 95

10.3 Battery connection

Caution ***Before connecting the battery sets, switch off the power supply by setting the main switch to "O".***

One battery set consists of two batteries connected in series at the factory. Connect the positive pole of battery set 1 first. Then connect the negative pole of battery set 1. The positive pole is marked with red and the negative pole with black. Proceed in the same way with battery set 2. See fig. 7.

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Fig. 7 Connecting the battery sets

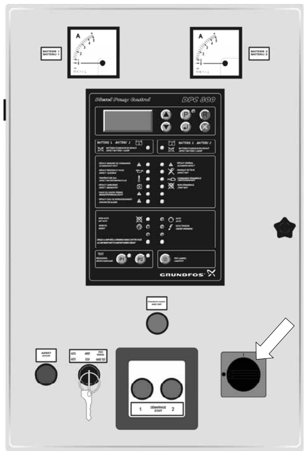
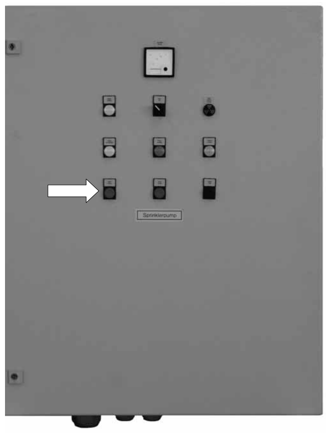
11. Startup

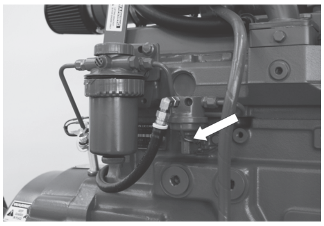
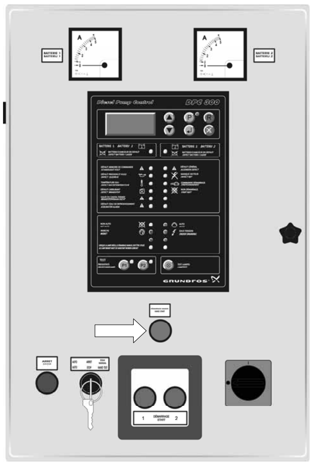
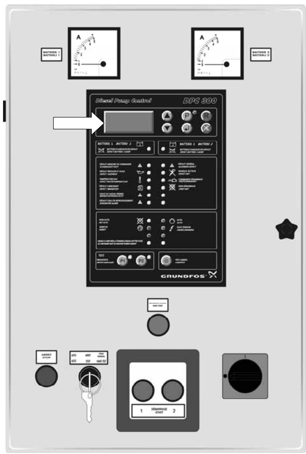
Warning
Startup must be carried out by authorised personnel. Do not touch any current-carrying parts when the cabinet door is open.

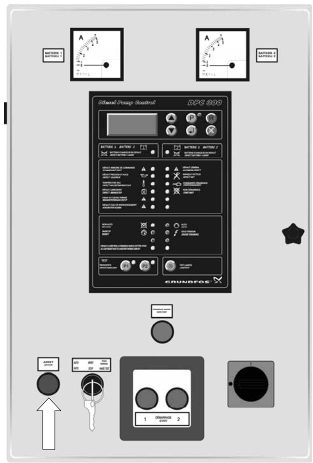
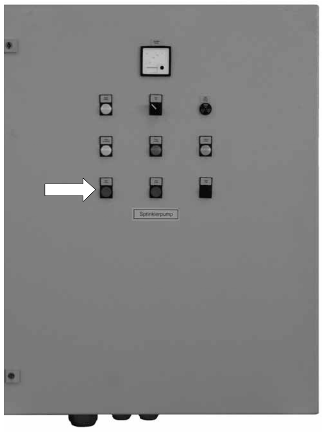
Warning
Observe the safety instructions and startup procedure in the installation and operating instructions for the diesel engine and the pump.

The following instructions presuppose that the controller has been installed according to section *10. Installation*, and that the pump set has been correctly connected to the controller.
The pump set must be prepared for startup according to the installation and operating instructions for the pump and diesel engine. Set the pressure switches to the pressure values required. See the installation and operating instructions for the pressure switches.

Note ***Before starting up the system, check that the terminals of the controller and signal transmitters are tight.***

Step	Illustration
1. Set the main switch to "I".	
2. Set the selector switch to "MANUAL TEST".	
3. Open the fuel cock on the fuel tank.	

Step	Illustration
4. Activate the manual fuel supply pump.	
5. Open the isolating valve on the suction side of the pump. Close the isolating valve on the discharge side of the pump. Open the isolating valve of the test pipe.	
6. Press [MANUAL START] until the engine starts, however, not for longer than 15 seconds per start attempt.	
Note: If the engine does not start, check the fault messages and instructions in the display. It may be necessary to vent the fuel system. See the installation and operating instructions for the diesel engine. For further causes, see section *15. Fault finding*.	
7. Slowly open the isolating valve on the discharge side of the pump. To vent, loosen the vent plug in the discharge line. As soon as liquid escapes, tighten the air vent screw to 25 Nm (1/2") or 30 Nm (3/8"). When the pump reaches operating pressure, open the isolating valve sufficiently to reach the duty point.	
8. Compare the oil pressure, temperature and speed in the display with the nominal values. See the data sheet for the diesel engine.	

Step	Illustration
9. Stop the diesel engine manually by pressing [STOP].	
10. Close the isolating valve of the test pipe.	
11. Set the selector switch to "AUTO".	

The pump set is now operational and in automatic mode.

Note

The controller is preset at the factory. Parameter values should only be changed if the engine does not start or is difficult to start.

In connection with the startup, a final test run must be carried out according to EN 12845 or ANPI guideline. To do this, activate the automatic startup command by lowering the pressure in the discharge pipe while the fuel valve is closed. Every starting cycle consists of a 5 to 10 second starting phase followed by a pause of maximum 10 seconds. After six failed start attempts, the start failure alarm must be activated. When the fuel valve has been opened and the alarm indication has been reset by pressing [R], the pump should start properly. See section *12.4 Test run*.

12. Operation

Warning

Observe the safety instructions in the installation and operating instructions for the diesel engine and the pump.

Caution ***Ensure sufficient ventilation of the controller!***

The most important operating parameters and alarm messages appear in the display and are indicated via indicator lights in the control panel. See section *8.3 Status and fault indications*. If the outputs of the pump set are connected to a building management system, the operation can be monitored remotely.

To remedy any faults arising, see section *15. Fault finding*. Alarms which are not automatically reset must be reset by pressing [R]. See fig. 8.

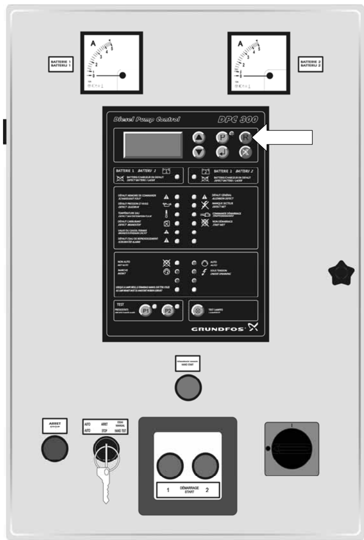


Fig. 8 Manual resetting of alarms

The four possible operating modes are described in the following subsections:

- Automatic operation (normal operation)
- Manual operation (for startup and after service work)
- Emergency operation (if automatic operation is disrupted)
- Test run.

12.1 Automatic operation

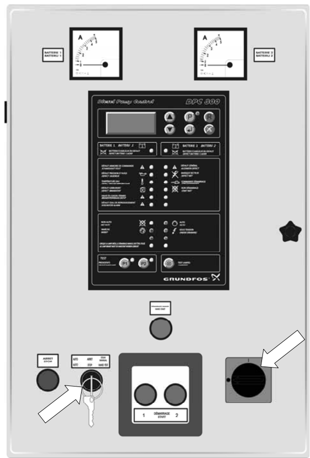
Once the pump set has been installed and started up according to the installation and operating instructions, no further operation is necessary.

The pump set works automatically and switches itself on as soon as water is drawn from the sprinkler system and the pressure switch detects a pressure drop.

Note ***In automatic mode, the selector switch must be set to "AUTO". The green "AUTO" indicator light will be on.***

System in automatic mode

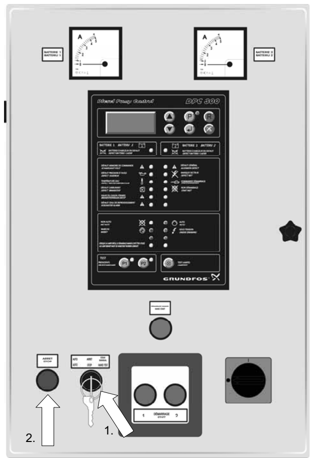
When the main switch is set to "I" and the selector switch is set to "AUTO", the pump set is in automatic mode.
Note: To secure the pump set against unauthorised use, remove the key.



Note ***The pump set is not stopped in case of warnings.***

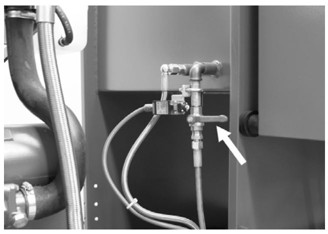
Stopping the system

The pump set is normally stopped by setting the selector switch to "MANUAL TEST" (first step) and pressing [STOP] (second step).

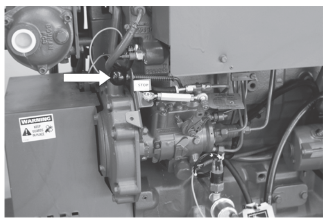


Emergency shutdown

Close the fuel cock on the tank for emergency shutdown.



Emergency shutdown is also possible by pulling the STOP lever on the injection pump. Alternatively, pull the STOP lever on the injection pump.



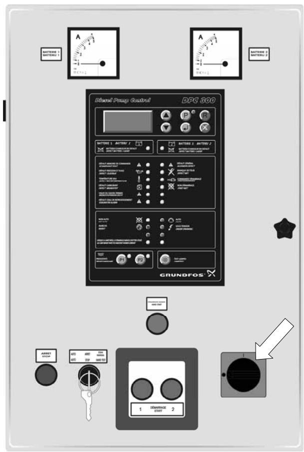
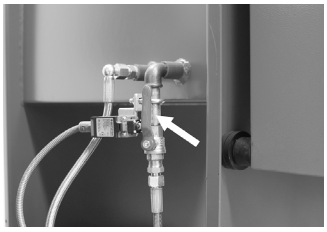
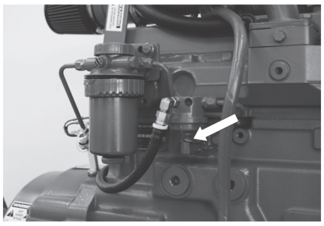
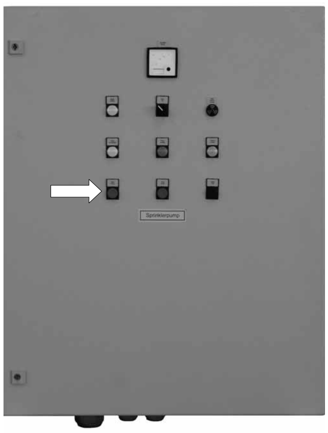
Note ***The main switch has no emergency shutdown function. Though the main switch is set to "O", the pump keeps running.***

12.2 Manual operation

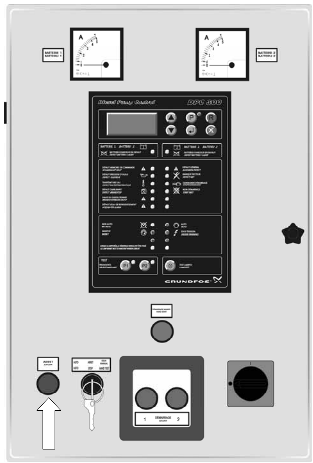
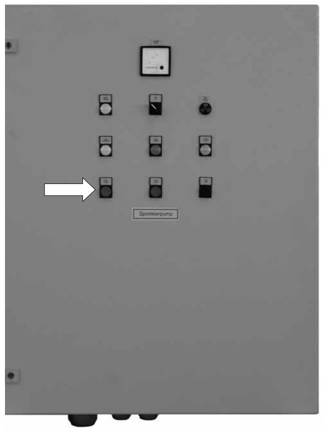
The pump set can also be manually started and stopped for a functional test, at startup or after service work.

Note	***The pump must be vented before startup.***

Starting procedure

Step	Illustration
1. Set the main switch to "I" if this has not already been done.	
2. Open the fuel cock on the fuel tank if this has not already been done.	
3. Activate the manual fuel supply pump.	
4. Set the selector switch to "MANUAL TEST" by means of the key.	
5. Open the isolating valve of the test pipe.	
Note: If the engine does not start, check the fault messages and instructions in the display. See section *15. Fault finding*.	

Stopping procedure

Step	Illustration
1. Stop the diesel engine manually by pressing [STOP].	
2. Close the isolating valve of the test pipe.	
3. Set the selector switch to "AUTO" by means of the key.	

Note	***After stopping, the pump set returns to automatic mode.***

12.3 Emergency operation

If the engine does not start after six start attempts in automatic mode, the automatic operation is blocked, and the yellow "START FAILURE" indicator light will light up in the control panel. The text "Start failure" will appear as a warning message in the display. Also, the yellow "OPERATE MANUAL START TEST BUTTON IF LAMP IS LIT" indicator light will light up. In this case, it is possible to start the fire pump set manually by means of [MANUAL START]. See fig. 9. Press the button until the engine starts, however, not for longer than 15 seconds. Stop the engine as described in section *12.1 Automatic operation*.

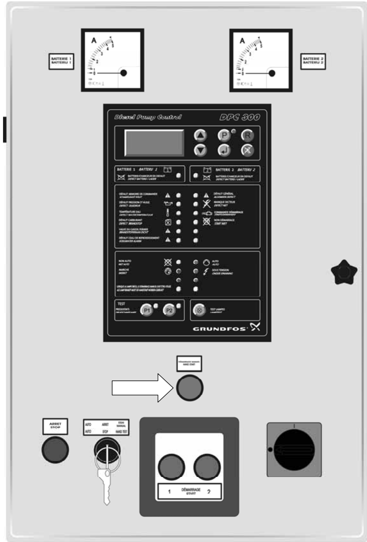


Fig. 9 Manual start via button after failed start attempts

If the pump set cannot be started with [MANUAL START], or if the controller is defective and, for example, the yellow "DIESEL CONTROLLER FAILURE" indicator light in the control panel is on, the last possibility is to start the engine via the two emergency start buttons behind the glass panel. See fig. 10. To do this, break the glass panel. Carry out the start attempts alternately using the left and the right button. Press the button until the engine starts, however, not for longer than 15 seconds. Stop the engine as described in section *12.1 Automatic operation*.

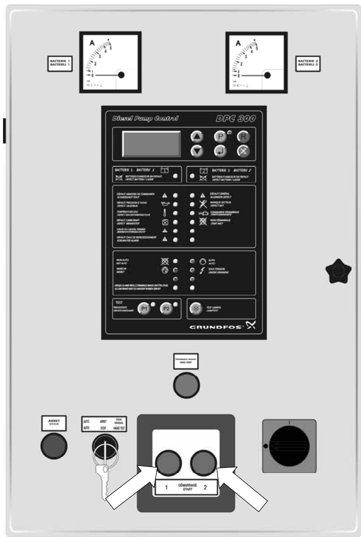


Fig. 10 Emergency start when controller fails

12.4 Test run

The function of the emergency start device and the automatic start function must be tested during a test run.

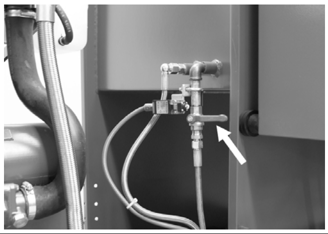
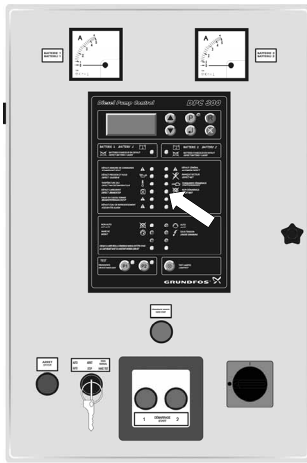
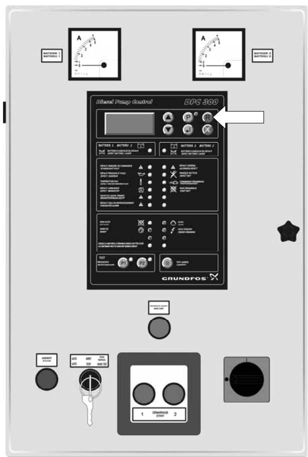
Caution

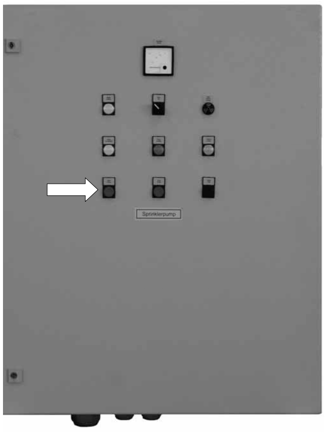
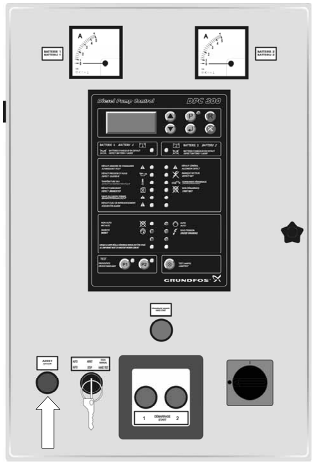
Do not leave the pump room during the test run. Observe all status and fault indications as the pump set does not stop automatically in case of failure (e.g. no cooling water or no oil).

Note

In the following description of the test run, it is assumed that the fire pump set was previously in automatic operation.

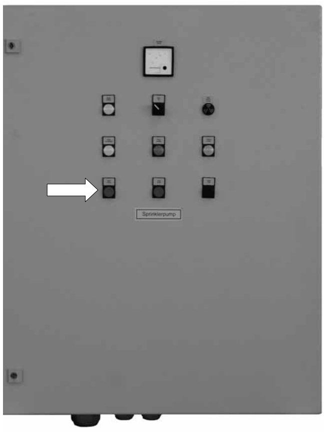
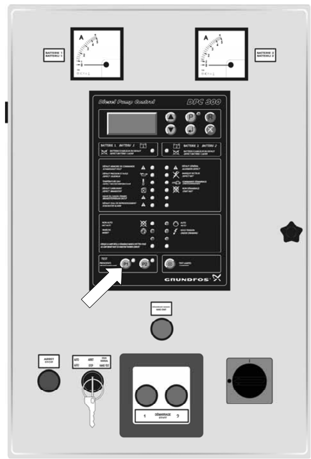
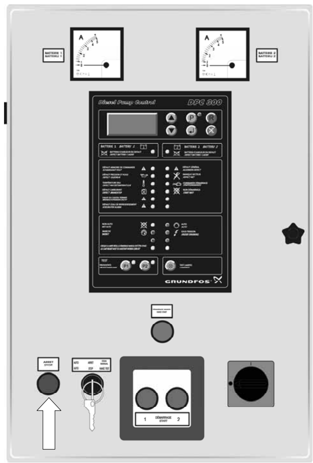
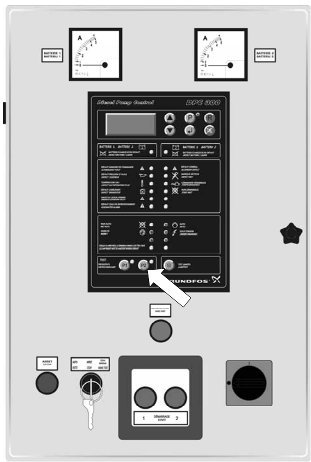
Testing the "FAILURE TO START" indicator light

Step	Illustration
1. Close the fuel valve on the fuel tank, and open the isolating valve of the test pipe. This results in a failed start attempt.	
2. After six failed start attempts, the "FAILURE TO START" indicator light lights up.	
3. Open the fuel valve on the fuel tank.	
4. Reset the alarm by pressing [R].	
5. Close the isolating valve of the test pipe.	

Step	Illustration
6. Set the selector switch to "MANUAL TEST" by means of the key.	
7. Stop the pump set by pressing [STOP].	
8. Set the selector switch to "AUTO".	

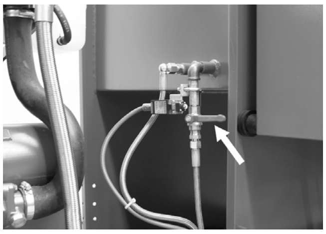
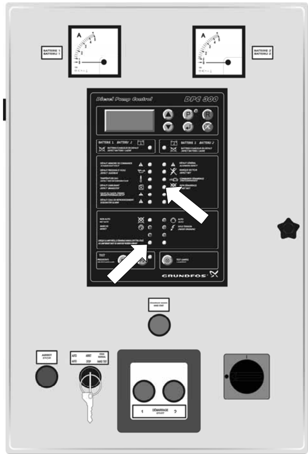
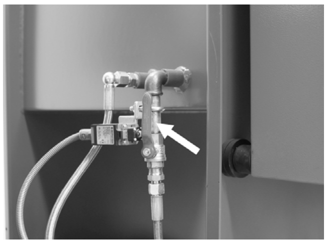
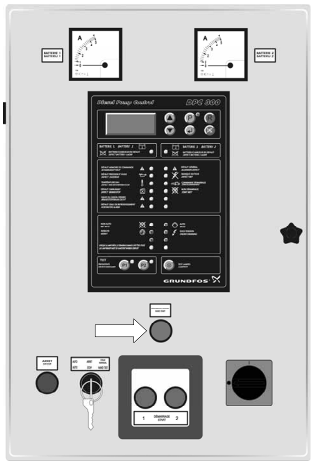
Note: ***After stopping, the pump returns to automatic mode.***

Testing the automatic start function

Step	Illustration
1. Set the selector switch to "MANUAL TEST" by means of the key.	
2. Open the isolating valve of the test pipe.	
3. Start the pump set by pressing [P1].	
4. Stop the pump set by pressing [STOP].	
5. Start the pump set by pressing [P2].	
6. Switch off the pump as described in section *12.2 Manual operation*.	

Note: ***After stopping, the pump returns to automatic mode.***

Testing the emergency startup device

Step	Illustration
1. Close the fuel valve on the fuel tank, and open the isolating valve of the test pipe. This results in a failed start attempt.	
2. After six failed start attempts the "START FAILURE" indicator light lights up together with the "OPERATE MANUAL START TEST BUTTON IF LAMP IS LIT" indicator light in the control panel.	
3. Open the fuel valve on the fuel tank.	
4. Start the pump set by pressing [MANUAL START] until the engine starts, however, not for longer than 15 seconds per start attempt.	

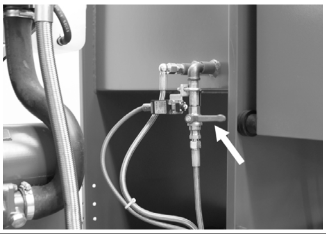
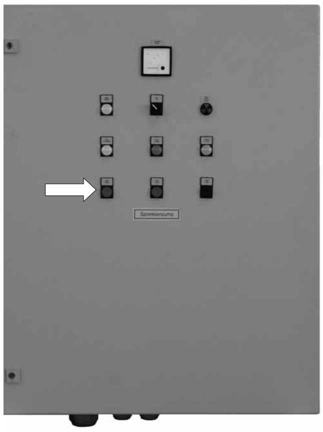
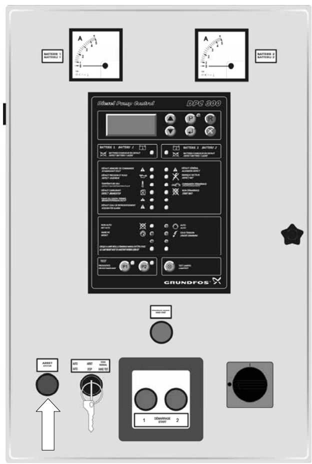
Step	Illustration
5. Stop the pump set by setting the selector switch to "MANUAL TEST" (first step) and pressing [STOP] (second step).	
6. Close the isolating valve of the test pipe.	
7. Set the selector switch to "AUTO".	

Note ***After stopping, the pump returns to automatic mode.***

13. Shutdown

Note — *If running, the pump should be stopped as described in section 12.2 Manual operation.*

Note — *See the installation and operating instructions for the pump and the diesel engine regarding draining of the pump and the fuel tank.*

Step	Illustration
1. Close the isolating valve on the discharge side.	
2. Close the isolating valve on the suction side.	
3. Turn off the fuel valve on the fuel tank.	
4. Set the selector switch to "STOP".	
5. Set the main switch to "O".	
6. Disconnect the negative pole and then the positive pole of each battery.	

14. Maintenance

Warning

Maintenance work must be carried out by authorised personnel.

Only trained personnel are allowed to open the cabinet door.

Switch off the power supply before carrying out any maintenance work.

Note — ***The operator is responsible for ensuring that all maintenance, inspection and installation work is performed by qualified personnel. A regular maintenance plan will help avoid expensive repairs and contribute to trouble-free, reliable operation.***

Note — ***Observe the regulations of the EN 12845 or ANPI guideline concerning maintenance.***

14.1 Controller

The following maintenance work must be carried out regularly:

- Test indicator lights once a week.
- Check wire connections once a year.

To test the indicator lights of the control panel, press [⊗]. See fig. 11. If an indicator light fails, please contact Grundfos.

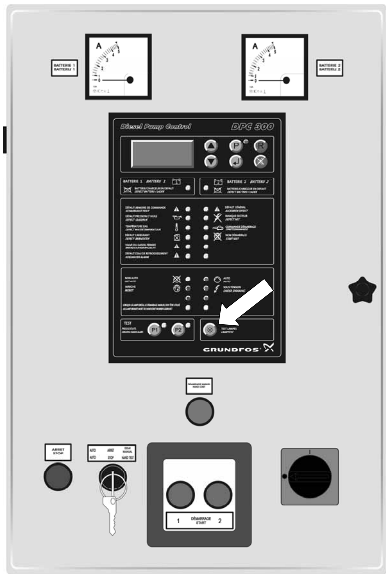


Fig. 11 Indicator light test

Once a year, check that all screw connections to the terminals and all earth connections are tight. Any loose connections must be tightened. Check the cables for visible damage and replace, if necessary.

14.2 Batteries

The batteries are maintenance-free. However, keep the batteries clean and dry to avoid leakage currents.

Plastic parts of the batteries, especially containers, must be cleaned with pure water without additives. Do not use organic solvents.

If the batteries have to be replaced, first remove the negative pole and then the positive pole. The positive pole is marked with red and the negative pole with black. Then remove the cable for connecting the two batteries in series. See section *10.3 Battery connection*.

15. Fault finding

Warning
Only trained personnel are allowed to open the cabinet door.

Warning
Service work must be carried out by authorised personnel.
Before servicing, make sure that the pump set cannot be accidentally started.

Note ***See the installation and operating instructions for the pump and the diesel engine.***

Fault	Possible cause	Remedy
1. No power supply to controller.	a) Main switch in position "O". No indicator lights on, and display not lit.	Switch on the controller by means of the main switch.
	b) Main fuse F1 has tripped.	Search for and remedy the cause. Cut in fuse F1.
	c) Fuse F6 has tripped.	Search for and remedy the cause. Cut in fuse F6.
	d) No mains voltage, or no battery connected. No indicator lights on, and display not lit.	Reestablish the power supply and connect the batteries. See sections *10.2 Electrical connection* and *10.3 Battery connection*.
	e) No power supply and no battery voltage due to defective batteries. "BATTERY/BATTERY CHARGER FAILURE" indicator lights on, and message "Fault, battery 1" or "Fault, battery 2" shown in display.	Check the cable and the mains connection. Check the battery capacity and the battery connection.
	f) No power supply and no battery voltage due to defective chargers. "BATTERY/BATTERY CHARGER FAILURE" indicator lights flashing, and message "Fault, battery 1" or "Fault, battery 2" shown in display.	Check the charger and the wires to the charger, and replace, if necessary.
	g) Fuse F2 or F3 for the charg2rs has tripped.	Search for and remedy the cause. Cut in fuse F2 or F3.
	h) Fuse F4 or F5 for the batteries has tripped.	Search for and remedy the cause. Cut in fuse F4 or F5.
2. Engine does not start in automatic mode despite power supply.	a) Selector switch not set to "AUTO". "AUTO" indicator light on.	Set the selector switch to "AUTO".
	b) Fuel valve closed. "VALVE CLOSED" indicator light on.	Open the fuel valve.
	c) No fuel in fuel tank. "FUEL STOCK" indicator light on, and message "Fuel reserve" shown in display.	Fill the fuel tank, and vent the fuel system, if necessary. See the installation and operating instructions for the diesel engine or the fuel tank.
	d) Both pressure switches or cable to pressure switch defective. Pressure switch incorrectly connected. Message "Alarm, press. 1" or "Alarm, press. 2" shown in display.	Test the pressure switches and replace, if necessary. Check the pressure switch connection according to *10.2 Electrical connection*.
	e) Duration of start attempts and time interval between start attempts not correctly set.	Change the settings. See sections *8.2.5 Start attempt interval* and *8.2.6 Duration of start attempts*.
	f) Defective controller. No indicator lights on, and display not lit.	In case of fire, break the glass and start the pump set with [START 1] or [START 2].
	g) Defective motherboard. "DIESEL CONTROLLER FAILURE" indicator light on.	In case of fire, break the glass and start the pump set with [START 1] or [START 2].

Fault		Possible cause	Remedy
3. Engine does not start in automatic mode after six start attempts.	a)	Possible causes as mentioned under 2b) to d). "START FAILURE" indicator light and "OPERATE MANUAL START TEST BUTTON IF LAMP IS LIT" indicator light on, and message "Start failure" shown in display.	Observe the warning indications in the display and via the indicator lights. Remedy the cause. Reset the warning indications requiring manual resetting by pressing [R]. If the engine still does not start automatically, start the pump set with the emergency start button.
	b)	Duration of start attempts and time interval between start attempts not correctly set. "START FAILURE" indicator light and "OPERATE MANUAL START TEST BUTTON IF LAMP IS LIT" indicator light on, and message "Start failure" shown in display.	Change the settings. See sections *8.2.5 Start attempt interval* and *8.2.6 Duration of start attempts*. Reset the warning message "Start failure" with [R]. If the engine still does not start automatically, start the pump set with the emergency start button.
4. Engine does not start in manual mode despite power supply.	a)	Fuel valve closed. "VALVE CLOSED" indicator light on.	Open the fuel valve.
	b)	No fuel in fuel tank. "FUEL STOCK" indicator light on, and message "Fuel reserve" shown in display.	Fill the fuel tank, and vent it, if necessary. See the installation and operating instructions for the diesel engine or the fuel tank.
5. Engine is running, but there is a warning indication.	a)	Oil pressure too low due to low oil level or incorrect oil (incorrect viscosity). "LOW OIL PRESSURE" indicator light on, and message "Low oil press." shown in display.	Add or change oil. See the installation and operating instructions for the diesel engine.
	b)	Engine temperature too high. "ENGINE OVERHEATED" indicator light on, and message "Engine temperat." shown in display.	Check the cooling water level and add cooling water, if necessary. See the installation and operating instructions for the diesel engine.
	c)	Insufficient battery voltage. One of the "BATTERY/BATTERY CHARGER FAILURE" indicator lights on, and message "Fault, battery 1" or "Fault, battery 2" shown in display.	Test the battery capacities. Replace the batteries, if defective. Check if fuse of battery 1 or 2 (F4 or F5) has tripped. Cut in, if necessary.
	d)	Insufficient charging voltage. One of the "BATTERY/BATTERY CHARGER FAILURE" indicator lights flashing, and message "Fault, charger 1" or "Fault, charger 2" shown in display.	Test the function of the chargers. Replace the chargers if they are defective. Check if fuse of charger 1 or 2 (F2 or F3) has tripped. Cut in, if necessary.
	e)	No power supply or low voltage. "SUPPLY ALARM" indicator light on, and message "Power supply" shown in display.	Check the power supply and re-establish it, if necessary.
	f)	Fuel shortage (level < 70 %). "FUEL STOCK" indicator light on, and message "Fuel reserve" shown in display.	Fill the tank with diesel. See the installation and operating instructions for the diesel engine or fuel tank.
	g)	Engine block heater. "COOLING WATER ALARM" indicator light on, and message "Short-c. heater" shown in display.	Check the function of the engine block heater. Replace the engine block heater if it is defective.
6. Engine continues running although [STOP] has been pressed.	a)	Stop valve defective.	Stop the engine by closing the fuel valve. Replace the stop valve.
	b)	Cable to stop valve defective or incorrectly connected.	Check the connection according to section *10.2 Electrical connection*. Test the cable and replace, if necessary.

16. Service, spare parts, accessories

Note ***Spare parts and accessories not supplied by Grundfos are not inspected or approved by Grundfos.***

The installation and/or use of such products may negatively alter and thus impair specified properties of the pump set and the controller.

The usage of non-authorised spare parts and accessories renders any liability on behalf of Grundfos for resulting damages null and void.

Any malfunctions which cannot be repaired should only be corrected by Grundfos or authorised specialist companies.

Please provide an exact description in the event of a malfunction so that our service technician can prepare and provide the appropriate spare parts.

Please obtain the technical data for the pump set or controller from the nameplate.

17. Warranty

The warranty is governed by the framework of our general terms of delivery. Liability for any damage which is a result of errors during installation, electrical connection or incorrect use is excluded. Liability for consequential damage is excluded.
The start of the warranty period is to be verified.

18. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the product Fire Control EN high diesel 12 & 24 V, to which this declaration relates, is in conformity with these Council directives on the approximation of the laws of the EC member states:

NL: EC overeenkomstigheidsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat het product Fire Control EN high diesel 12 & 24 V, waarop deze verklaring betrekking heeft, in overeenstemming is met de Richtlijnen van de Raad in zake de onderlinge aanpassing van de wetgeving van de EG lidstaten betreffende:

— Machinery Directive (2006/42/EC)
  Standards used: EN 60204-1:2006
— Low Voltage Directive (2006/95/EC)
  Standards used: EN 61439-1:2010
— EMC Directive (2004/108/EC)
  Standards used: EN 61000-6-2:2006, EN 61000-6-3:2007

Bjerringbro, 03th July 2014

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Person authorised to compile technical file and
empowered to sign the EC declaration of conformity.

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